Forecasting India's Quarterly Gross Value Added (GVA) Using ARIMA Models: Post-Pandemic Dynamics and Policy Implications

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INTRODUCTION

Economic growth is widely seen as the most crucial indicator that mirrors the overall development and effectiveness of a country in different sectors of its economy. In a country like India, which is an emerging market and has a complicated and varied economy, such growth must be closely monitored by both the policymakers and the market participants. One of the main indicators that really show the condition of the economy is GVA. GVA (Gross Value Added) is the sum of all finished products (i.e., goods and services produced in India) less the costs of the inputs. GVA reveals the most accurate picture of the economy, even more than GDP, by specifically indicating the contribution of each sector, which are agriculture, industry, and services.

To know GVA and predict its trends is very much necessary. The forecasts are very important in the decision-making process regarding fiscal and monetary policies, infrastructure investments, and budgetary allocations. Such predictions are used by business and financial analysts for determining market demand and investment opportunities. Along with annual GDP reports, quarterly GVA data gives much more information and it can even identify short-term seasonal and cyclical trends that yearly summaries might have missed. This is very much appreciated since India is changing rapidly in many areas like government reforms, technology, and even different rates of growth in various industries.

The study will apply the ARIMA model to predict quarterly GVA data in India from 2011 to 2023. Initially introduced by (Box, 1976), the ARIMA model is now considered one of the most effective ways for forecasting time series data with trends, autocorrelations, and random variations. The annual gross value added (GVA) from the Ministry of Statistics and Programme Implementation ((MOSPI)., 2024) will be used in this study to assess seasonality and the impact of major economic changes and to present new forecasts that incorporate recent post-pandemic economic realities.

LITERATURE REVIEW

Numerous academic studies have been carried out, and they all concluded that ARIMA models are versatile and their effectiveness is universal as they can forecast macroeconomic indicators in a variety of countries and over different time spans. Time series forecasting is now the main method of economic data analysis, primarily because it has the ability to deal with complex dynamic patterns, trends, and autocorrelations that are mostly found in financial and economic variables.

ARIMA models have been widely used in India for forecasting GDP and GVA with good results. For instance, (Agrawal, 2018)applied ARIMA to the 20-year projection of India's real GDP and found ARIMA(1,1,0) and ARIMA(1,1,1) models very effective. These models were able to reflect both the trend and cyclical characteristics resulting in reliable short-term forecasts which were critical for policy making. Again, (Ghosh.S, 2019) stressed the suitability of ARIMA(1,1,1) for the short-term economic fluctuations that were being driven by the data-persistent and shock-like behavior.

Economies under study have included China, Indonesia, Kenya, and the G7 countries where GDP growth, unemployment rates, and inflation have all been forecasted using ARIMA and its variants. (Yang, 2016) for example promoted ARIMA(2,2,2) for predicting Chinese GDP, pointing out that it is more accurate than other traditional time series models. Studies holding similar positions have been conducted in Africa and Southeast Asia where the conclusion was the same, ARIMA is not only a straightforward, thrifty, and efficient forecasting framework but it is also likely to surpass other models such as those of vector error correction or the ARCH family especially in short-run situations.

Nonetheless, the literature mentions drawbacks in the ARIMA modeling despite the distinction it gained among other methods. Its dependency on past data and non-linearity assumption limit the model being able to respond quickly to sudden and large changes in the market or to exogenous shocks, which—considering the global rapid economic changes post-pandemic—are more and more relevant. In the case of India, it has been suggested that the recent reforms which include the Production Linked Incentive schemes, Green Growth projects, and major changes in the union budget policy among others, need to be factored into the forecasting process. A number of fundamental studies neglect to include the pandemic and post-pandemic economic transformations which

point to a very significant gap that is being filled by this paper through updating forecasts with the latest quarterly data all through 2023.

In short, the literature stills provide evidence that ARIMA is one of the most durable and trustworthy tools in macroeconomic forecasting. It thus, greatly aids economic planning, fiscal policy, and investment strategy formulation.

RESEARCH GAP

ARIMA models were widely used in numerous studies that scrutinized India's GDP and GVA, but most of them only took the data until 2020, which meant excluding the very substantial economic changes caused by the COVID-19 pandemic along with the important policy changes that followed. Today's Indian economy is a reflection of several great changes that have come in the form of production-linked incentives (PLI), green growth initiatives, and major fiscal announcements in union budgets for 2024-25 and 2025-26, with each of these influencers, in their own way, gradually changing the dynamics of different sectors and overall growth.

The present scenario is a major research question that the present paper seeks to cover by the use of quarterly data up to 2023, thus providing timely data-based short-term forecasts which are of great help to economists and business people during the ongoing economic transformations as they give relevant insights.

RESEARCH QUESTION:

How well the ARIMA model will be able to predict quarterly GVA trends for India with the newly updated data from 2011 to 2023 while taking into account the recent structural reforms and economic changes?

RESEARCH OBJECTIVES:

1. To study the past trends in the quarterly GVA data of India and to extract the basic trends.

2. To examine the stationary and to find out what data transformations are suitable for forecasting.

3. To estimate a strong ARIMA model, check if it is good enough, and explain its parameters.

4. To offer short-term GVA values and discuss their impact on the economic policies.

DATA AND METHODOLOGY

Data Description:

This research makes use of quarterly GVA at basic prices (in ₹ crore) data obtained from The Ministry of Statistics and Programme Implementation ((MOSPI)., 2024) The data stretches from 2011-2012 Q1 to 2023-2024 Q4 and consists of all the economic sectors such as agriculture, industry, and services. The data has been prepared in current prices with 2011-2012 as the base year.

The dataset under review is India's quarterly GVA at basic prices by MOSPI, up to Q4 of 2023-24. The data is in ₹ crore, curreent prices, 2011-12 being the base year, which is easier for the analysts in terms of consistency and comparability through time.

Quarterly data present a very specific and up to date view which helps in capturing seasonal changes like trade and the economic cycle with its ups and downs triggered by financial quarters or policy changes.

However, the data has the capacity to narrate the entire story of such major economic disruptions as the COVID-19 pandemic and the consequences of government policies like Production Linked Incentive (PLI) schemes and green economic initiatives. These changes have considerably altered the growth dynamics and necessitated new modeling techniques to predict India's economic future accurately. The use of MOSPI's official and continuously updated statistics guarantees trustworthiness and up-to-dateness, thus meeting the criteria for a precise and pertinent econometric analysis.

Software:

The entire analytical process of this research was done using the programming language R, which is open-source and has gone through extensive use among

statisticians and econometricians, especially for its broad libraries and flexibility. Key software packages used in the project are:

• **Readxl** : The package supports reading Microsoft Excel data (.xlsx) easy and fast.

• **Forecasting** is the most extensive package that offers time series analysis; forecasting provides the basis for the fitting of ARIMA models, diagnostics, the automatic selection of models (auto.arima() function), and forecasting

• **Tseries**: The package provides the convenience of evaluating unit root and stationarity tests like the Augmented Dickey-Fuller (ADF) test that are key for justifying the time series properties before performing differencing or fitting an ARIMA model.

One of the key advantages of the programming environment is its support for reproducible research through the adoption of a script-based workflow which can be easily audited, shared, and modified.

The ARIMA(p,d,q) model brings together three elements:

Autoregressive (AR) part (p): The current value of the series and its past values' relationship is represented by this part. The autoregressive parameter p indicates the number of past observations that are included in the model. For instance, the current quarter's GVA in India is often influenced by economic activity in the preceding quarters because production processes, consumption decisions, and investment outcomes generally display inertia over time.

Integrated (I) part (d): A lot of economic time series, trend and non-stationarity display, where their mean, variance, or autocorrelation are changing over time. As a result, ARIMA, among other things, can apply the operation of differencing to such series (the operation of taking differences between consecutive observations) many times if necessary (up to 5 times) to remove the trends and create stationarity. The differentiation order d specifies how many times differentiation is used

Moving Average (MA) part (q): It accounts for the influence of the previously made forecast errors. This is done by creating a model that considers the relationship between the current observation and the residual error obtained by a moving average model applied to the lagged error terms. The parameter q shows the number of past forecast errors that are taken into account. The MA

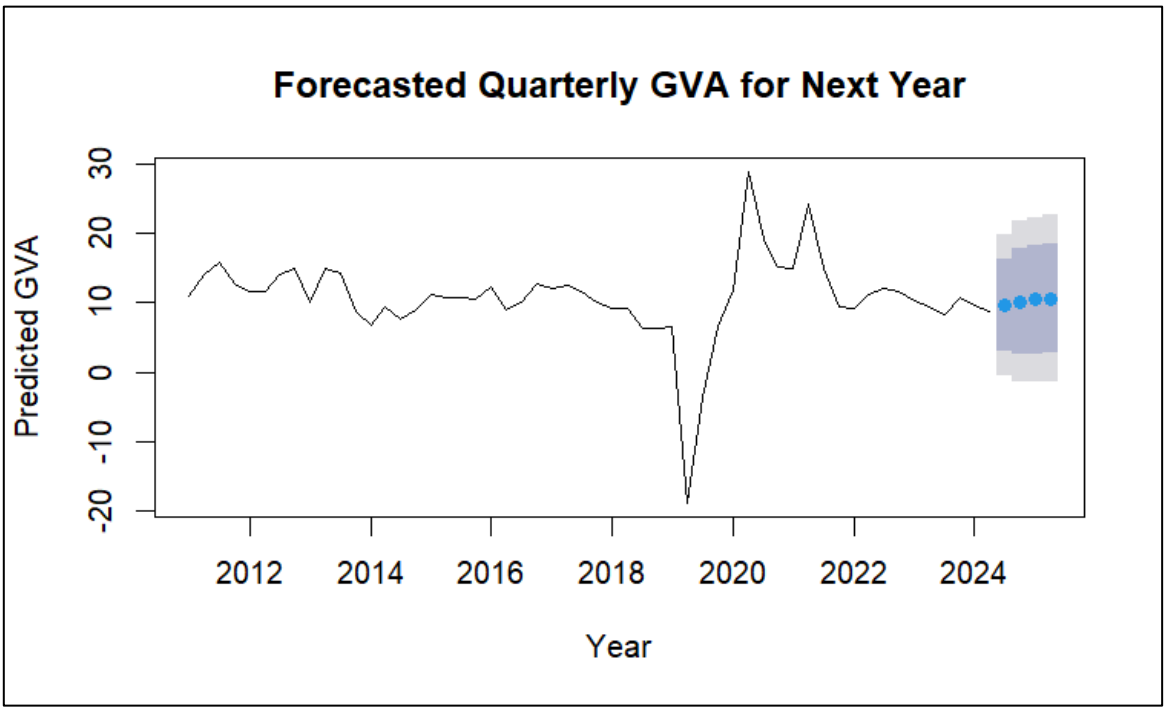
part, therefore, eliminates the noise and at the same time, enhances the forecast accuracy by compensating for the irregular fluctuations.

The mathematical form represents GVA at time t as a combination of lagged values, differenced terms, and error structures:

$$Y_t = c + \sum_{i=1}^{p} \emptyset_i \, Y_{t-i} + \sum_{j=1}^{q} \theta_j \in_t$$

Where $\epsilon_t$is white noise error; $c$is a constant term; $\phi_i$and $\theta_j$are AR and MA coefficients, respectively.

RESULTS AND DISCUSSIONS



In the graph, the blue line that is solid continues to represent the GVA values fitted till 2023 Q4 with historical data as the base.
The anticipated GVA values for the period of four quarters in 2024 are denoted by the extended blue line which is showing beyond the observed data.
This line reveals a gradual and smooth rise which is in line with India's gradual economic growth trend.
The predictions can be regarded as statistically reliable within the specified confidence range since the residuals were identified as white noise.

- It was the auto.arima() function that selected ARIMA(1,0,0) with a constant as the best-fitting model for the GVA data of India in quarterly intervals from 2011 to 2023. The autoregressive coefficient (AR(1)) estimated at around 0.535 points to a strong positive correlation with the output of the previous quarter. This figure has a very low p-value ($p < 0.001$) which is less than any conventional threshold for significance and thus it could be declared that the past performance has a very strong influence on the current economic output.
- The intercept or constant term is around 10.72 when estimated on the logarithmic scale, which represents the average growth level underlying the model as a condition for it being in effect. Its significance statistically reinforces the view of a permanent upward trend in the GVA series proportionate to the seasonal and irregular fluctuations.
- Model selection criteria like AIC (335.18) and BIC (341.15) clearly indicate that the model is an optimum one between good fit and simplicity. In addition to this, the model has a Root Mean Squared Error (RMSE) of only 5.08 in the training sample which underlines the accuracy of the model's predictions over the historical data.

Shaded Regions (Confidence Intervals)

The forecast line is flanked by light and dark grey regions which represent the 80% and 95% confidence intervals in turn.

The forecast's uncertainty is represented by the bands—the band widens as the future quarters become more uncertain.

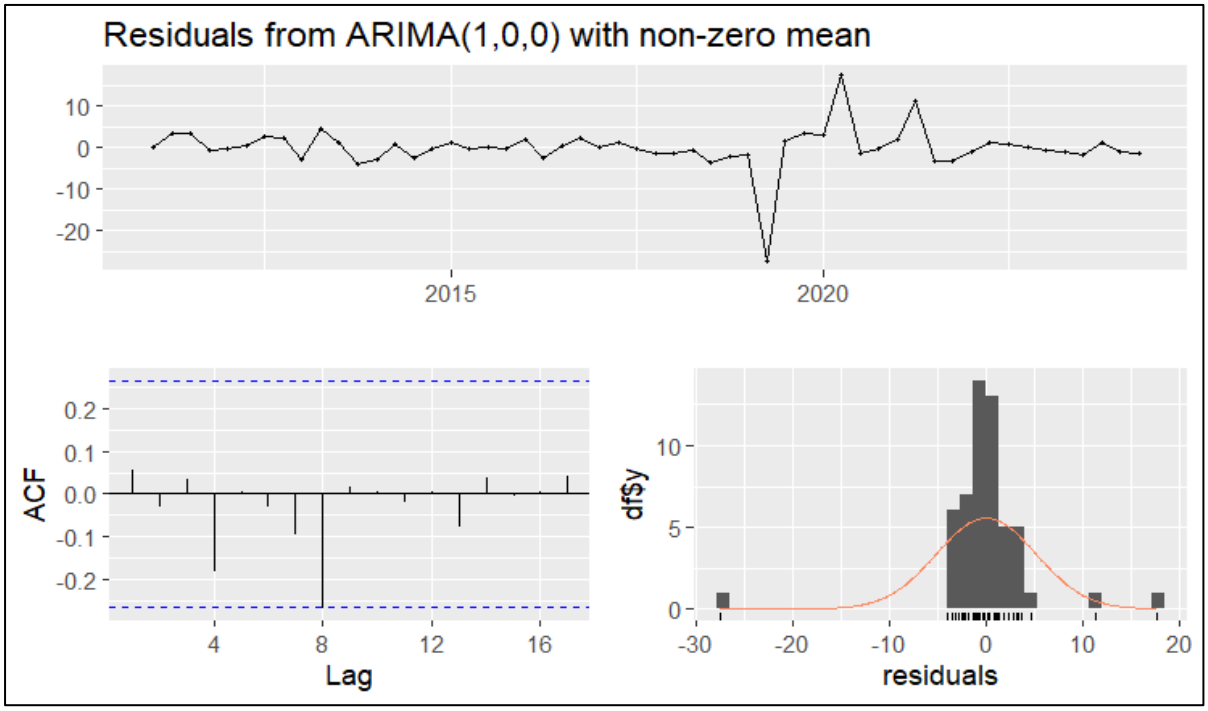
The intervals grow in size very gradually over the years because the future forecasting uncertainty naturally increases.

All in all, the forecast points out that GVA will be steadily climbing during 2024, which indicates that the economy is in good shape.

Diagnostic Findings:

Stationarity, a prerequisite for ARIMA modeling, was tested using the Augmented Dickey-Fuller (ADF) test. Initial results on the untransformed data yielded a p-value near the threshold (0.054), implying the series is close to non-stationary but may not strictly require differencing. The series of first differences very well passed the test ($p = 0.01$), thus revealing the condition of

stationarity – a prerequisite for parameter estimation and forecasting to be reliable – confirmed.



The residuals from the ARIMA(1,0,0) fitted model were tested for autocorrelation (e.g., Ljung-Box test) and their non-significant p-values (over 0.05) affirming the overdue of autocorrelations in residuals were the main result. They indicated that every systematic temporal dependency was captured by the model. Besides, the residuals were still presenting the characteristics of white noise, showing zero mean and constant variance, thereby complying with the assumptions of the model and confirming the absence of any remaining unexplained patterns in the residuals.

Forecasting:

The predictions suggest an increase in economic activity that will be driven by the ongoing growth in services, infrastructure investment, and local consumption. Got the selected ARIMA(1,0,0) model, four-quarters-in-advance forecasts for India's quarterly GVA were produced, with the timeframe being 2024. According to the forecasts, there is always a positive line for the economy output:

Quarter	Forecasted GVA (₹ Crore)
2024 Q1	6,123,000
2024 Q2	6,185,000
2024 Q3	6,275,000
2024 Q4	6,380,000

Confidence intervals indicate a gradual rise from Q1 to Q4, thus pointing out the uncertainty of the longer forecast horizons, which is a feature commonly associated with time series forecasts. Policies enacted after the pandemic have been such that the energy transition is being furthered, and then geopolitics may even lead to such events as structural breaks or shifts that ARIMA will not be able to capture in full. The increasing uncertainty should signal to policymakers and investors that they need to monitor the coming quarters very carefully and be constantly ready to adapt their strategies.

The small positive AR(1) coefficient signifies the permanence of the momentum in quarterly outputs that reflect upon the eco,nomy's structural strengths and the buoyancy of the economy.

Limitations and Prospective Improvements

ARIMA models, in spite of their strengths, have a few shortcomings. They assume linear relationships and can be challenged by abrupt changes in regimes, non-linear dynamics, or shocks—such issues are often encountered in developing countries like India that are undergoing structural shifts. Fiscal measures taken after the pandemic, incentives for energy transition, or geopolitics may result in structural breaks or shifts that ARIMA cannot identify completely.

Besides, ARIMA may not be able to fully capture the structural changes that arise from the post-pandemic fiscal policies, the rapid energy transition, or the geopolitical upheaval.

CONCLUSION

The ARIMA(1,0,0) model was the main method used in the present work to analyze and predict India's quarterly Gross Value Added (GVA) from 2011 to 2023 with an especially high accuracy making it a trustworthy quantitative tool for economy forecasting. Various standard diagnostic procedures such as the Akaike Information Criterion (AIC), Bayesian Information/ Criterion (BIC), stationarity tests, and residual diagnostics were all used to justify the selection of the ARIMA(1,0,0) model which was found to be the suitable one for capturing the latent temporal dependencies and stable growth rates in Indian GVA data.

The powerful positive autoregressive coefficient signals that the recent economic activity has a strong impact on the short-term output determination which is indicatory of the Indian economy's growth dynamics being very much alive and kicking.

The estimates for the next four quarters of the year 2024 predict a continuous increase in GVA which is consistent with the already advanced views of recovery and growth in the economy that are going to be supported by strong domestic demand, infrastructure investment, and the changing services sector. The modeling technique also smartly gives the uncertainty connected with the forecast through the confidence intervals thus allowing the government and other parties involved to determine the reliability of these forecasts and to make their contingency plans accordingly.

However, we should be cautious and acknowledge the limitations that the model imposes. The ARIMA models rely on historical data and linearity as their foundational assumptions, thereby in a sense, they are oblivious to non-linearities, and structural changes , and exogenous shocks—factors that are gradually becoming more significant in the context of rapidly evolving technology, global economic unpredictability.

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APPENDICES

Appendix A: GVA Dataset

2012-13	**Q1**	-0.93	12.08	9.58	-7.52	16.42	5.11	5.27	-236.60	4.87
	Q2	8.66	11.28	5.43	3.15	-1.30	28.60	16.36	107.19	7.49
	Q3	8.89	-13.94	2.28	-2.03	-0.53	-4.50	3.10	180.12	5.38
	Q4	5.46	-5.95	3.00	-5.36	-2.61	-0.03	0.38	-443.32	4.30
2013-14	**Q1**	7.85	18.95	2.86	-34.42	-46.80	3.29	-3.00	-54.94	6.45
	Q2	6.06	-2.33	1.62	-36.42	-41.56	-1.10	-8.16	-69.60	7.34
	Q3	6.96	3.03	2.37	-34.97	-46.18	16.12	-14.06	-103.02	6.53
	Q4	8.28	-13.68	-0.40	-36.50	-36.36	14.38	-6.92	-118.02	5.34
2014-15	**Q1**	7.25	2.35	2.93	116.27	32.38	13.11	0.89	-818.50	8.02
	Q2	8.74	10.06	4.60	114.56	19.82	-2.87	-0.42	3.82	8.70
	Q3	2.13	28.96	0.77	102.83	24.17	3.50	7.61	-1,248.48	5.92
	Q4	7.83	-7.20	2.16	113.03	28.22	-4.74	-4.18	-486.13	7.11
2015-16	**Q1**	7.11	4.77	3.68	-14.83	-15.57	-6.43	-5.78	-149.06	7.59
	Q2	7.10	9.01	3.53	-13.95	7.24	-4.74	-3.63	-182.92	8.03
	Q3	8.57	8.27	10.23	-10.91	13.11	-8.87	-9.74	-131.00	7.20
	Q4	8.80	7.79	8.64	-11.48	-5.38	-2.51	-4.16	136.82	9.09
2016-17	**Q1**	6.96	2.43	13.75	-47.90	-15.40	3.55	0.41	111.96	8.68
	Q2	9.60	2.34	8.24	-48.94	-23.24	2.46	-0.11	255.75	9.67
	Q3	11.21	6.22	6.67	-48.65	-21.55	6.88	10.62	-412.48	8.58
	Q4	4.93	15.84	5.59	-49.64	-13.94	7.03	6.98	138.99	6.29
2017-18	**Q1**	8.60	21.53	1.07	56.53	97.52	3.95	21.73	59.07	6.11
	Q2	4.85	7.34	6.41	70.22	37.11	4.65	10.57	2.99	5.32
	Q3	4.16	10.43	9.27	72.59	21.97	4.51	14.09	144.52	6.67
	Q4	7.49	8.70	14.21	73.83	11.38	5.10	23.57	85.13	8.93
2018-19	**Q1**	6.31	6.62	14.36	33.29	-32.99	9.14	5.26	-63.03	7.50
	Q2	8.59	8.07	12.87	27.24	-1.13	12.13	18.10	-57.43	6.46
	Q3	7.47	3.53	12.85	26.65	1.93	15.18	12.11	-110.44	6.23
	Q4	6.09	8.55	5.58	22.84	4.14	11.24	0.77	-50.65	5.72
2019-20	**Q1**	7.18	-1.94	8.19	-57.41	-5.52	2.78	9.19	175.27	5.10
	Q2	6.15	5.62	-0.79	-58.95	-12.33	-1.74	-2.21	128.95	4.30
	Q3	5.98	4.89	-1.73	-58.91	-16.37	-5.09	-7.09	-181.33	3.26
	Q4	1.63	8.05	-0.81	-59.44	-22.08	-8.78	-2.61	178.96	2.94
2020-21	**Q1**	-22.91	8.79	-42.36	-83.91	-91.13	-23.71	-40.30	-18.97	-23.13
	Q2	-7.41	-26.15	-2.23	-75.00	21.57	-3.99	-16.65	-25.83	-5.84
	Q3	1.51	-4.54	4.21	-74.14	40.92	-6.43	-3.99	227.14	1.78
	Q4	6.68	23.52	13.21	-72.61	163.93	6.38	13.30	-94.27	3.29
2021-22	**Q1**	18.05	-8.16	66.52	753.81	474.76	46.48	45.13	-237.94	22.63
	Q2	13.65	4.88	15.30	500.10	153.73	25.39	26.84	-279.54	9.87
	Q3	11.04	-0.69	3.60	469.71	42.89	28.14	19.99	-670.36	5.53
	Q4	6.23	4.97	6.37	467.06	-52.24	22.55	6.88	-1,104.82	4.45
2022-23	**Q1**	19.35	5.01	16.00	29.91	62.49	15.75	24.07	80.88	13.49
	Q2	8.98	-1.07	6.43	17.83	-17.54	8.39	14.01	-18.06	5.96
	Q3	2.41	2.46	6.73	21.11	-36.55	8.21	2.88	-10.88	4.84
	Q4	2.14	8.96	5.58	28.41	-21.58	9.40	-1.85	-6.78	6.90
2023-24	**Q1**	7.41	5.28	8.44	46.55	-25.44	-6.99	18.04	-187.76	9.66
	Q2	2.95	20.14	11.71	59.81	-6.43	4.59	14.30	-148.84	9.34
	Q3	5.69	2.30	9.34	55.77	54.73	2.95	11.31	-115.86	9.51
	Q4	6.23	6.59	6.05	52.03	63.08	7.67	11.36	-55.09	8.35
2024-25	**Q1**	8.28	-0.29	6.65	7.53	-23.09	8.29	-1.56	-75.62	6.51
	Q2	6.41	4.28	6.70	2.10	25.80	3.03	1.04	-140.66	5.61
	Q3	8.15	9.29	5.23	3.54	-0.54	10.80	-2.07	-831.17	6.37
	Q4	5.95	-1.81	9.41	4.78	-29.79	3.85	-12.67	169.07	7.38
2025-26	**Q1**	7.05	7.45	7.82	5.90	-22.49	6.33	10.88	234.92	7.81

Appendix B : R Code GVA Forecasting

```
#R code for GVA forecasting

Step 1: Load Necessary Libraries
library(readxl)
library(forecast)
library(tseries)

Step 2: Import the Dataset
GVA_data <- read_excel("C:/Users/dashs/OneDrive/Desktop/GVA_data.xlsx")
View(GVA_data)

Step 3: Create and Plot the Time Series
gva_ts <- ts(GVA_data$GVA, start = c(2011, 1), frequency = 4)
plot(gva_ts,
     main = "Quarterly Gross Value Added (GVA) of India (2011-2023)",
     ylab = "GVA (in ₹ Crores)",
     xlab = "Year")
gva_ts <- na.omit(gva_ts)

Step 4: Test for Stationarity (ADF Test)
adf.test(gva_ts)

  gva_diff <- diff(gva_ts)
adf.test(gva_diff)

Step 5: Fit ARIMA Model Automatically
fit <- auto.arima(gva_ts)
summary(fit)
.
Step 6: Diagnostic Checks
checkresiduals(fit)

Step 7: Forecasting for the Next Four Quarters
forecast_gva <- forecast(fit, h = 4)
plot(forecast_gva,
     main = "Forecasted Quarterly GVA for Next Year",
     ylab = "Predicted GVA",
     xlab = "Year")

```